26 Two charged particles, X and Y, are accelerated from rest through the same potential difference V. They then enter a region of uniform magnetic field of magnetic flux density B acting perpendicular to their velocity.

Particle X has mass m and charge q, and moves in a circular path of radius r.
Particle Y has mass $2m$ and charge $4q$.

What is the radius of the circular path of particle Y?

A $\frac{r}{2}$ **B** $\frac{r}{\sqrt{2}}$ **C** r **D** $\sqrt{2}r$